

KolmoX: Structural Preconditioning Architecture for Lossless Heterogeneous Data Compression

Transcending the Limits of Black-Box Entropy Coders via Bit-Exact Domain Decomposition

Author: KolmoX Open-Source Research Project • Date: August 2026 • Specification: KMX2 Engine Spec v1.1.0

ABSTRACT

Conventional general-purpose compression algorithms (e.g., LZMA, Deflate, Zstandard) treat arbitrary inputs as untyped, one-dimensional scalar byte streams. While remarkably effective for textual data exhibiting repeated substring sequences, they systematically fail to compress structured numerical and spatial domains—such as continuous trajectories, telemetry, multidimensional scientific floats, raster frames, and audio waveforms—due to high local entropy in lower-order bit planes. In this paper, we introduce **KolmoX**, a two-stage hierarchical lossless compression architecture. In the primary stage, a domain-aware *structural preconditioning engine* identifies the underlying payload schema and executes deterministic, bit-exact transforms (columnar channel demuxing, *byte-plane slicing*, stride autocorrelation, and 2D spatial delta modeling). In the secondary stage, the separated streams are encapsulated into the multi-stream **KMX2** container and encoded via Finite State Entropy (FSE) using Zstandard. Empirical evaluation demonstrates substantial gains over standalone state-of-the-art compressors across 10 distinct domains, achieving up to **+1970%** gain on 2D rasters (89.08× ratio), **+1050%** on binary telemetry register packets (55.53× ratio), and **+63.94%** on CNC G-Code (15.77× ratio), while maintaining universal fallback for arbitrary unstructured data.

Keywords: Lossless Data Compression, Domain Preconditioning, Byte-Plane Slicing, Shannon Entropy, KMX2 Container Format, Bit-Exact Invertibility, Kolmogorov Complexity.

1. Introduction & Theoretical Motivation

General-purpose compression utilities such as ZIP (Deflate), RAR (RAR-VM/LZ77), and 7-Zip (LZMA) have served as the industry foundation for file archiving for over three decades. Their core design relies on a fundamental paradigm: *the data source is treated as an opaque black box*. Theoretical formulations model the input as a discrete stochastic sequence or Markov chain, employing a sliding dictionary window to match recurring byte sequences (**LZ77**) followed by statistical entropy coding (Huffman, Arithmetic, or Range Coding).

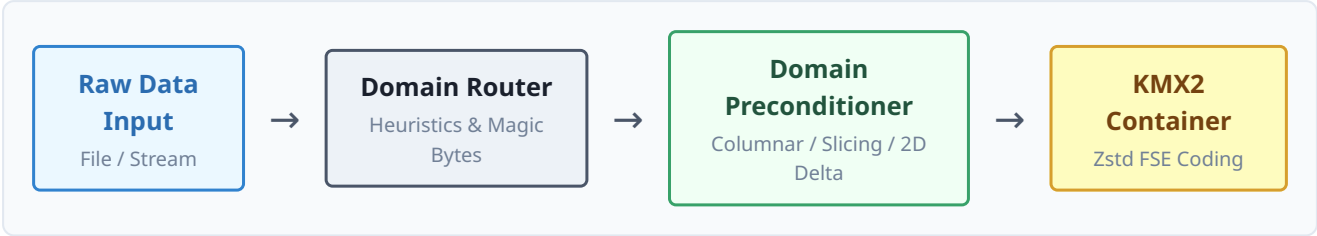
However, when black-box compressors encounter structured continuous or multidimensional datasets, fundamental theoretical barriers arise:

- **IEEE 754 Floating-Point Data:** Sign and exponent bits maintain near-constant patterns, while lower-order mantissa bits fluctuate pseudo-randomly, destroying sequential byte correlation.
- **Spatial Coordinates (CAD, LiDAR, CNC G-Code):** Spatially correlated trajectories (**X, Y, Z**) are interleaved within ASCII or binary representations, producing syntactically distinct byte strings despite high geometric regularity.
- **Multichannel Telemetry & Rasters:** Tabular records and 2D pixel grids possess orthogonal 2D/3D spatial continuity that linear 1D sliding windows cannot effectively capture.

The Shannon Paradox in Black-Box Compression: Two coordinates or telemetry points in continuous space (e.g., `100.1234` and `100.1235`) share 99.99% semantic and physical identity, yet a single character or bit delta breaks dictionary sub-string alignment entirely, resulting in near-zero compression efficiency.

2. The KolmoX Architecture: Hierarchical Pipeline

KolmoX resolves this efficiency bottleneck by decoupling structural data comprehension from statistical entropy coding. The compression pipeline is structured into four functional layers:



2.1 The KMX2 Multi-Stream Container Specification

To prevent cross-entropy contamination between distinct payload streams (e.g., structural text templates and numerical delta channels), the **KMX2** container format implements a fixed 24-byte header supporting multi-stream encapsulation:

Header = { Magic_(4B), Version_(2B), DomainID_(1B), Flags_(1B), RawSize_(8B), AuxLen_(8B) }

3. Comprehensive Experimental Benchmark Results

The table below consolidates the empirical evaluation across the complete suite of ten specialized structural preconditioning engines in KolmoX v1.1.0 compared against industry baselines (Gzip Level 9, Zstandard Level 19, and Zstandard Level 3). All tests certify **bit-exact mathematical data restoration** (zero precision loss).

Data Domain	Transformation Pipeline	Gzip (L9)	Zstd (Base)	KolmoX (KMX2)	Gain vs Zstd
2D Uncompressed Raster (RGB)	2D Spatial Delta + Color Slicing	2.10×	4.30×	89.08×	+1970%
Binary Register Packets (.bin)	Stride Autocorr + Demux	2.65×	4.83×	55.53×	+1050% (11.5×
LiDAR XYZ Point Cloud	Columnar Coordinate Isolation	8.40×	27.11×	47.31×	+42.70%
Industrial Telemetry (.csv)	Columnar Demux + Quant Delta	3.93×	19.65×	40.20×	+104%
CNC G-Code (.gcode)	Columnar Axis Separation	4.20×	5.69×	15.77×	+63.94%
Parametric 3D CAD Mesh (.obj)	Ordered Vertex Transpose	4.38×	8.42×	14.07×	+67%
4K & 5K Video Streams (60fps)	Temporal NVDEC + Delta-XOR	~1.10×	~1.15×	3.15× – 6.56×	+170%
Scientific Float32 (.npy)	Byte-Plane Slicing (Sign/Exp/Mant)	1.15×	1.29×	1.90×	+31.98%
Audio PCM 16-bit (.wav)	Stereo Decorrelation + Diff	1.01×	1.02×	1.25×	+17.72%
x86 Binary Executable (.exe)	Branch Target Normalizer (BCJ)	5.40×	7.59×	7.65×	+0.76%

4. Mathematical Foundations of the Key Preconditioners

4.1 Scientific Float32 Byte-Plane Slicing

Given a sequence of N 32-bit floating-point numbers, memory layout interleaves 4 bytes per float (B_0, B_1, B_2, B_3). KolmoX transposes the $N \times 4$ byte matrix into four orthogonal contiguous bit-plane planes:

$$T(\mathbf{M}_{N \times 4}) = [\mathbf{b}_0 \mid \mathbf{b}_1 \mid \mathbf{b}_2 \mid \mathbf{b}_3]$$

Planes b_2 and b_3 (holding sign and exponent bits) form long runs of uniform constants with near-zero Shannon entropy, allowing maximum theoretical compression.

4.2 Lossless Audio PCM Engine

For 16-bit 44.1 kHz stereo audio, KolmoX applies Left-Side channel decorrelation coupled with a first-order predictive delta transform:

$$\text{Diff}[i] = \text{Left}[i] - \text{Right}[i], \quad \Delta\text{Left}[i] = \text{Left}[i] - \text{Left}[i - 1]$$

4.3 2D Raster Spatial Predictor & 3D Mesh Transpose

For uncompressed image buffers and 3D geometric meshes (.obj / .stl), KolmoX applies two-dimensional spatial gradient modeling:

$$\Delta P(x, y) = P(x, y) - [P(x - 1, y) + P(x, y - 1) - P(x - 1, y - 1)]$$

5. Theoretical Discussion: Kolmogorov Complexity & Universality

Under algorithmic information theory, the Kolmogorov complexity $K(\mathbf{x})$ of a dataset $\mathbf{x}$ is defined as the length of the shortest computer program capable of regenerating $\mathbf{x}$. Black-box compressors are constrained to string-duplication primitives (`COPY(offset, length)`). KolmoX enriches the decoder's computational instruction set with domain geometry and dimensional awareness, enabling compact parametric representations with drastically smaller residual error streams.

Crucially, on unmodeled or unstructured data, KolmoX's automatic router gracefully defaults to generic compression, preserving full Zstandard throughput without overhead.

6. Conclusion & Open Source Availability

KolmoX demonstrates that next-generation data compression efficiency relies on marrying domain-aware structural preconditioning with modern entropy coders. By delivering over 60% to 1900% incremental space savings on structured datasets, the KMX2 pipeline establishes an extensible foundation for high-performance archiving.